Hospital Data Analysis Questions

Based on 12 linked tables: patients, doctors, departments, appointments, admissions, diagnoses, medications, prescriptions, lab tests, billing, insurance and staff.

Patients and Demographics

1. How many patients are registered, and how has registration grown by year?
2. What is the age and gender distribution of patients?
3. Which cities have the most patients?
4. What percentage of patients have insurance, and which providers are most common?

Appointments

5. What is the completion, cancellation and no-show rate?
6. What are the most common cancellation reasons?
7. Which days and time slots are busiest?
8. What is the average satisfaction score by doctor and by department?
9. Which doctors have the highest cancellation rates?

Admissions

10. What is the average length of stay by department and room type?
11. What is the readmission rate, and which departments or diagnoses drive it?
12. What is the breakdown of admission status (discharged, transferred, etc.)?
13. Are some admissions missing a discharge date, and why?
14. How does bed occupancy compare with each department's num_beds?

Diagnoses and Clinical

15. What are the top 10 diagnoses, and how does severity vary across them?
16. Which diagnoses are most linked to readmissions or long stays?
17. Which lab tests are ordered most, and what share of results are abnormal?
18. Which medications are prescribed most, and does usage differ by diagnosis?
19. What is the average prescription duration by medication?

Doctors and Staff

20. How does doctor experience relate to consultation fee and satisfaction?
21. Which specialties have the most doctors per department?
22. What is the monthly salary cost by role and by department?
23. How does staff salary cost compare with each department's operating cost?

Billing and Finance

24. What is the total revenue by month, department and year?
25. How much is paid, partially paid, pending or overdue?
26. What is the average bill for appointments versus admissions?
27. How much does insurance cover versus patients pay out of pocket?
28. What is the average delay between bill date and payment date?
29. Which patients or departments have the highest unpaid balances?

Insurance

30 Which insurers cover the most patients and have the highest coverage amounts?
31 How many policies have expired or are expiring soon?
32 Do patients with insurance have higher bills or more admissions?

Data Quality

33 Which columns have missing values (cancellation reasons, results, discharge dates)?
34 Are there duplicate records or orphan foreign keys, such as bills pointing to missing appointments?
35 Do any dates look inconsistent, for example discharge before admission or a prescription before admission?